

Chapter 1: Number Systems

1.1 Introduction In your earlier classes, you have learnt about the number line and how to represent various types of numbers on it. You may have also learnt about irrational numbers and how to locate them on the number line.

A number is called a rational number if it can be written in the form p/q , where p and q are integers and q is not equal to 0. For example, $1/2$, $3/4$, and $-5/7$ are all rational numbers.

An irrational number is a number that cannot be written in the form p/q for any integers p and q . Examples include the square root of 2, π , and e .

The collection of all rational and irrational numbers is called real numbers. Every real number is represented by a unique point on the number line. Also, every point on the number line represents a unique real number.

1.2 Irrational Numbers We saw in the previous section that there may be numbers on the number line that are not rationals. In this section, we are going to investigate these numbers. The Pythagoreans in ancient Greece discovered that all numbers cannot be expressed as ratios of integers.

The square root of 2 is irrational. This was proved by the Greek mathematicians. If the square root of 2 is rational, then we can find integers a and b where b is not zero such that the square root of 2 equals a/b . We can assume that a and b have no common factors. Then a^2 squared equals 2 times b^2 squared. This means a^2 squared is divisible by 2, so a itself must be divisible by 2. Writing a as $2c$, we get $4c^2$ squared equals $2b^2$ squared, which gives b^2 squared equals $2c^2$ squared. This means b is also divisible by 2, contradicting our assumption.

1.3 Real Numbers and their Decimal Expansions In this section, we are going to study rational and irrational numbers from a different

point of view. We will look at the decimal expansions of real numbers and see if we can use them to distinguish between rationals and irrationals.

The decimal expansion of a rational number is either terminating or non-terminating recurring. For example: - $1/2 = 0.5$ (terminating) - $1/3 = 0.333\ldots$ (non-terminating recurring) - $1/7 = 0.142857142857\ldots$ (non-terminating recurring)

The decimal expansion of an irrational number is non-terminating non-recurring. For example, the square root of 2 = 1.41421356... where no block of digits repeats.

1.4 Operations on Real Numbers We have learnt how to add, subtract, multiply and divide rational numbers. We can also perform these operations on real numbers.

The sum or difference of a rational number and an irrational number is irrational. The product or quotient of a non-zero rational number with an irrational number is irrational.

For example: $2 + \text{square root of } 3$ is irrational, and $7 \text{ times the square root of } 5$ is irrational.

1.5 Laws of Exponents for Real Numbers We have studied the laws of exponents in earlier classes. The same laws hold for real number exponents: - $a^m \text{ times } a^n = a^{(m+n)}$ - $(a^m)^n = a^{(mn)}$ - $a^m / a^n = a^{(m-n)}$ - $a^m \text{ times } b^m = (ab)^m$